

Table of contents

Electro-Magnetic Field (force caused by potential):

$$\mathcal{B} = \nabla \times U_m = \frac{1}{c} \left(\frac{r}{|r|} \times \mathcal{E} \right) \quad \begin{array}{c} \text{Near Field} \\ \approx \end{array} \dots \begin{array}{c} \text{Far Field} \\ \approx \end{array} \dots \quad (1)$$

$$\mathcal{E} = -\nabla U_e - \frac{\partial}{\partial t} U_m \quad \begin{array}{c} \text{moving charge} \\ = \end{array} \frac{1}{2} \frac{1}{2\pi} \frac{1}{\epsilon_0} \frac{1}{r} \frac{1}{\left(1 - \frac{r}{|r|} \cdot \frac{v}{c}\right)^3} q \left[\underbrace{\left(1 - \frac{v^2}{c^2}\right) \frac{\frac{r}{|r|} - \frac{v}{c}}{r}}_{\text{uniform movement}} + \underbrace{\frac{\frac{r}{|r|} \times \left[\left(\frac{r}{|r|} - \frac{v}{c}\right) \times \dot{v}\right]}{c^2}}_{\text{accelerated movement}} \right] \quad \begin{array}{c} \text{fast moving cl} \\ = \end{array} \quad (2)$$